This assignment will not be graded and is only for practice.

Key Points: 1

Definition of a vector space V:

A vector space V over $\mathbb{R}$ is a set along with two functions:

$+ : V \times V \rightarrow V$ (V is closed under vector addition i.e., if $x, y \in V \implies x + y \in V, \ \forall x, y \in V$)
$. : \mathbb{R} \times V \rightarrow V$ (V is closed under scalar multiplication i.e., if $x \in V$ and $c \in \mathbb{R} \implies c.x \in V, \ \forall x \in V, \text{ and } \forall c \in \mathbb{R}$)
Axioms of vector additions of a vector space
$v_1 + v_2 = v_2 + v_1, \ \forall v_1, v_2 \in V.$
$v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3, \ \forall v_1, v_2 \text{ and } v_3 \in V.$
There exists an element 0 (called the zero vector of V) in V such that $0 + v = v, \ \forall v \in V$.
There exists a vector v' in V such that $v' + v = v + v' = 0, \ \forall v \in V$ (v' is called the inverse (or negative) of v with respect to addition).

Axioms relating vector addition and scalar multiplication of a vector space (since this is usual multiplication in real numbers, we suppress the "." sign).
For each vector $v \in V$, $1v = v$.
For each vector $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(a + b)v = av + bv$.
For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.
For each vector $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(ab)v = a(bv)$.

Consider a set $V = \{(1, -y) \mid y \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the usual addition and scalar multiplication as in $\mathbb{R}^2$. Answer the following questions (1) and (2).

1) Which of the following option(s) is (are) true? **1 point**

[**Hint:** Check the axioms of vector space.]

- [ ] $(1,2) \in V$.
- [ ] $(2,3) \in V$.
- [ ] $V = \mathbb{R}^2$.
- [ ] $v_1 = (1,-5)$ and $v_2 = (1,7) \in V \implies v_1 + v_2 \in V$.
- [ ] $v = (1,9) \in V, c \in \mathbb{R} \implies cv \in V, \forall c \in \mathbb{R}$.
- [ ] For all $v \in V, cv \in V$ only for $c = 1$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(1,2) \in V$.
For all $v \in V, cv \in V$ only for $c = 1$.

2) Which of the following option(s) is (are) true? **1 point**

- [ ] Zero element does not exist in V.
- [ ] Zero element of V is (0,0).
- [ ] Inverse element of (1, 5) is (-1, -5).
- [ ] Inverse element of (1, 5) is (1, -5).
- [ ] Inverse element of (1, 5) does not exist.
- [ ] Inverse element of any element of V does not exist.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Zero element does not exist in V.
Inverse element of (1, 5) does not exist.
Inverse element of any element of V does not exist.

Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^3$ with the usual addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$,
where $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ and
scalar multiplication: $c(x, y, z) = (cx, cy, cy)$, $c \in \mathbb{R}$ and $(x, y, z) \in V$.
Answer the following questions (3) and (4).

3) Which of the following option(s) is (are) true? ***1 point***

[Hint: For a vector $v = (x, y, z)$, check whether the condition $x + y + z = 0$ is satisfied or not.]

- [ ] (1, 2, -3) $\in V$.
- [ ] (-1, 2, 1) $\in V$.
- [ ] $V = \mathrm{R}^3$.
- [ ] $(5, -7, 2) \in V$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(1, 2, -3) $\in V$.
$(5, -7, 2) \in V$.

4) Inverse element of (-2, -4, 6) with respect to vector addition is ***1 point***

- ( ) (2, 4, -6).
- ( ) (2, -4, -6).
- ( ) (2, 4, 6).
- ( ) (-2, 4, -6).

No, the answer is incorrect.
Score: 0

Accepted Answers:

(2, 4, -6).

5) Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^3$ with the usual addition as in $\mathbb{R}^3$ and scalar multiplication is defined as ***1 point***

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}, cz) & c = 0 \end{cases} \qquad (x, y, z) \in V,\ c \in \mathbb{R}$$

Which of the following option(s) is (are) true?

[**Hint:** By adding any two elements in V, or by multiplying a scalar with any element in V, check whether the resultant element is in V or not.]

- [ ] V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
- [ ] V is closed under scalar multiplication i.e., $(c \in \mathbb{R}, v \in V \implies cv \in V)$.
- [ ] V has zero element with respect to addition.
- [ ] $1.v = v$, where $1 \in \mathbb{R}$ and $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V is closed under addition i.e., $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
V has zero element with respect to addition.

6) Consider usual addition and scalar multiplication on $\mathbb{R}^2$ and the subset $V = \{(1, -y) \mid y \in \mathbb{R}\} \subseteq \mathbb{R}^2$. ***1 point***

Which of the following option(s) is (are) true?

[**Hint:** Take two arbitrary elements $(1, -y_1)$ and $(1, -y_2)$ from V and check the conditions given in the options.]

- [ ] For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(a + b)v = av + bv$.
- [ ] For each $a \in \mathbb{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.
- [ ] For each element $v \in V$ and for each pair $a, b \in \mathbb{R}$, $(ab)v = a(bv)$.
- [ ] The set V with the usual addition and scalar multiplication in $\mathbb{R}^2$ is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each element $v \in V$ and for each pair $a, b \in \mathrm{R}$, $(a + b)v = av + bv$.
For each $a \in \mathrm{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$.
For each element $v \in V$ and for each pair $a, b \in \mathrm{R}$, $(ab)v = a(bv)$.

7) Consider a set $V = \{(x, y, z) \mid x + y + z = 0, x, y, z \in \mathrm{R}\} \subseteq \mathrm{R}^3$ with the usual addition as in R^3 and scalar multiplication is defined as **1 point**

$$c(x, y, z) = \begin{cases} (0, 0, 0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}, cz) & c = 0 \end{cases} \qquad (x, y, z) \in V, \; c \in \mathrm{R}$$

Which of the following option(s) is (are) true?

- ○ For each element $v \in V$ and for each pair $a, b \in \mathrm{R}$, $(a + b)v = av + bv$
- ○ For each $a \in \mathrm{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$
- ○ For each element $v \in V$ and for each pair $a, b \in \mathrm{R}$, $(ab)v = a(bv)$
- ○ V is a vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

For each $a \in \mathrm{R}$ and for each pair $v_1, v_2 \in V$, $a(v_1 + v_2) = av_1 + av_2$

Key Points: 2

Definition of a subspace: A subset W of a vector space V is a subspace of V if it is a vector space with respect to the same operations defined on V.
Criteria to check a subset W of a vector space V is a subspace:
Zero vector should be in W i.e., $0 \in W$.
If two vectors v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
Let $c \in \mathrm{R}$ and if $v \in W$, then $cv \in W$

8) Let $W = \{(x, y) \mid x + 3y = 1, x, y \in \mathbb{R}\} \subseteq \mathbb{R}^2$. Which of the following options are true? ***1 point***

- [ ] Zero element exists in W.
- [ ] If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.
- [ ] Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
- [ ] W is not a subspace of $\mathbb{R}^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \notin W$.
W is not a subspace of $\mathbb{R}^2$.

9) Let $W = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$ be a subset of a vector space $M_{2\times2}(\mathbb{R})$. Which of the following option(s) is (are) true? ***1 point***

- [ ] Zero vector does not exist in W.
- [ ] If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
- [ ] Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
- [ ] W is a subspace of $M_{2\times2}(\mathbb{R})$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and $v_2 \in W$, then $v_1 + v_2 \in W$.
Let $c \in \mathbb{R}$ and if $v \in W$, then $cv \in W$
W is a subspace of $M_{2\times2}(\mathbb{R})$.

Key Points: 3

Linear combination: Let V be a vector space and $v_1, v_2, ...v_k \in V$. The linear combination of $v_1, v_2, ...v_k$ with coefficients $a_1, a_2, ...a_k \in \mathrm{R}$ is the vector $v = \sum_{i=1}^{k} a_i v_i$.
A vector $v \in V$ is said to be a linear combination of $v_1, v_2, ...v_k \in V$ if there exist $a_1, a_2, ...a_k \in \mathrm{R}$ so that $v = \sum_{i=1}^{k} a_i v_i$

10) Which of the following vectors are linear combinations of (2, 1) and (6, 3) with usual addition and scalar multiplication in $\mathbb{R}^2$? **1 point**

- [ ] (2, 1) + 5(6, 3)
- [ ] (2, 1)
- [ ] (6, 3)
- [ ] (3, 6)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(2, 1) + 5(6, 3)
(2, 1)
(6, 3)

11) Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ *i.e.*, $v_1 = av_2 + bv_3 + cv_4$, then which of the following options are true? **1 point**

- [ ] If $a = 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.
- [ ] If $a = 0$ and $b = 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.
- [ ] v_2 is a linear combination of v_2.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $a = 0$, then v_2 is a linear combination of $v_i, i = 1, 3, 4$.
If $a = 0$ and $b = 0$, then v_3 is a linear combination of $v_i, i = 1, 4$.
v_2 is a linear combination of v_2.

Your score is: 0/11